

Modeling of an Algae Pond

Octave Litrico

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Introduction

In the context of modeling an algae cultivation pond, the evolution of a scalar quantity $\rho(x, t)$, representing the concentration of algae can be described by a partial differential equation of the advection-diffusion type:

$$\frac{\partial \rho}{\partial t} + a \frac{\partial \rho}{\partial x} = \mu \frac{\partial^2 \rho}{\partial x^2}$$

where:

- a is the advection (transport) coefficient related to the movement induced by the water wheel
- μ is the diffusion coefficient (natural or turbulent dispersion).

However, to simplify the numerical analysis and better understand diffusion phenomena alone, we choose in this work to neglect the advection term. This amounts to considering only a pure diffusion equation:

$$\frac{\partial \rho}{\partial t} = \mu \frac{\partial^2 \rho}{\partial x^2}$$

1 The discretized Model

1.1 Discretization principle

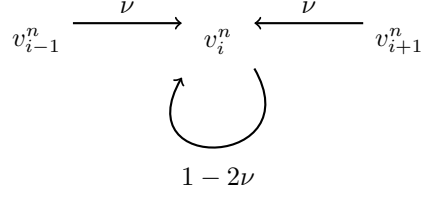
Let us discretize the time and space intervals. Let $N, M \in \mathbb{N}$, $h = \Delta_x = \frac{L}{N}$, $\Delta_t = \frac{T}{M}$ and $x_i = i\Delta_x$, $t_n = n\Delta_t \forall i \in \{0, \dots, N\}, n \in \{0, \dots, M\}$.

The pond will be modeled by a rectangle of size $L \times H$ divided into I horizontal layers of same size. The pond is circular, therefore the concentration functions of each layers will have to be periodic with period L . A water wheel located on the x -coordinate 0 stirs the water and will be modeled by a matrix P . It will intervene in the computing of the values of the algae concentration near 0. So that there is no production nor loss of matter, P must be column-stochastic. Indeed, we have that for all $i, j \in \{1, \dots, I\}$, $P_{i,j}$ is the amount of matter in line j sent to line i . Therefore, we must have that for all $j \in \{1, \dots, I\}$, $\sum_{k=1}^I P_{k,j} = 1$

The concentration function ρ will be an element of $\mathcal{C}^1([0, T], H^1([0, L])^I)$. At each time $t_n = n\Delta_t$, the concentration in cell j is denoted $\rho_j^n = \rho(t_n, x_j)$.

1.2 Cellular automat

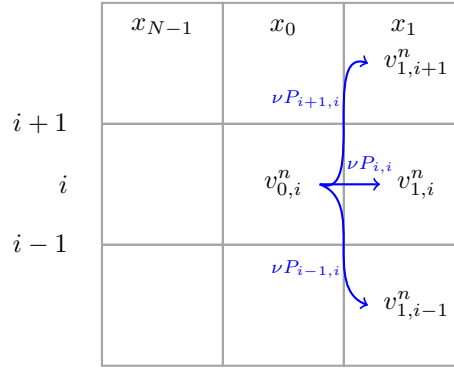
To first understand the underlying motion of the pond, let us introduce a cellular model. With the discretization showed above, we consider a cell in which the algae concentration evolves as a particle. A cell located far from the paddle wheel exchanges mass with its two neighbors. The scheme can be represented as follows:



Here, the value at time $n + 1$ is a weighted linear combination of the cell and its neighbors. ν here represents the mass proportion that each cell exchanges.

1.3 Special case: near the paddle wheel

Near the paddle wheel, the stir influences the concentrations. The following scheme represents the evolution, with the wheel located in column 0 :



Therefore we end up with this automat :

$$\begin{aligned}
& \forall n \in \{0, \dots, M-1\} \\
& \begin{cases} v_j^{n+1} = \nu(v_{j-1}^n + v_{j+1}^n) + (1-2\nu)v_j^n & \forall j \in \{2, \dots, N-2\} \\ v_{N-1}^{n+1} = \nu(v_{N-2}^n + P v_0^n) + (1-2\nu)v_{N-1}^n \\ v_0^{n+1} = \nu(v_{N-1}^n + v_1^n) + (1-2\nu)P v_0^n \\ v_1^{n+1} = \nu(P v_0^n + v_2^n) + (1-2\nu)v_1^n \end{cases}
\end{aligned}$$

2 PDE analysis

Let $\rho(t, x) \in \mathbb{R}^I$ be the vector of algae concentration on the x-coordinate of the pond at time t. We have to take into account the L -periodicity :

$$\rho(t, 0) = \rho(t, L) \quad \forall t$$

Furthermore, the total algae mass in the pond is constant through time. Therefore, we have :

$$\sum_i \int_0^L \rho_i = 0$$

up to a translation of the values of the ρ_i by the average concentration.

Let us denote U the set :

$$\{v \in H^2([0, L])^I \mid v(0) = v(L) = P v(L) \text{ and } \sum_{i=1}^I (\partial_x v_i(L) - \partial_x v_i(0)) = 0\}$$

and

$$V \text{ an eigen vector of } P \text{ such that } P V = V \text{ and } \|V\|_1 = 1$$

The ponctual values of $\partial_x u$ are well defined. Indeed, thanks to the Morrey injection, we have that $H^2([0, L]) \hookrightarrow \mathcal{C}^1([0, L])$. U is a Hilbert space for the following inner product :

$$\langle u, v \rangle_{U,U} = \sum_{i=1}^I \frac{1}{V_i} \int_0^L \partial_x u_i \cdot \partial_x v_i$$

The problem we are looking to solve is :

$$\begin{cases} \text{Find } \rho \in U \\ \text{such that } \partial_t \rho = \mu \Delta \rho \end{cases}$$

Firstly, let us notice that :

$$\begin{aligned} \forall v \in U, \\ \langle -\Delta \rho, v \rangle_{U,U} &= - \sum_{i=1}^I \frac{1}{V_i} \int_0^L \partial_{xx} \rho_i v \\ &= - \sum_{i=1}^I \frac{1}{V_i} \left[(\partial_x \rho_i(0) v_i(0) - \partial_x \rho_i(L) v_i(L)) - \int_0^L \partial_x \rho_i \partial_x v_i \right] \\ &= - \sum_{i=1}^I \left(\partial_x \rho_i(0) - \partial_x \rho_i(L) - \int_0^L \partial_x \rho_i \partial_x v_i \right) \quad \text{because } v_i(0) = v_i(L) = V_i \\ &= \sum_{i=1}^I \int_0^L \partial_x \rho_i \partial_x v_i \end{aligned}$$

We would like to find a Hilbert basis of U which will ease our difficulties. Let us introduce $H = \{h \in L^2([0, L])^I \mid \sum_i \int_0^L h_i = 0\}$, we can easily show that U is dense in H and by using the Rellich theorem, it follows that the embedding of U in H is compact.

Let us define for all $u, v \in U$:

$$a(u, v) = \mu \sum_{i=1}^I \int_0^L \partial_x u_i \cdot \partial_x v_i$$

The operator a is symmetric, bilinear, and coercive on U by the Poincaré inequality. Moreover, it is continuous. Therefore, there exists an infinite sequence $(\lambda_n)_{n \in \mathbb{N}} \in \mathbb{N}$ of positive eigenvalues of this operator, and a Hilbert basis $(x \mapsto \rho_n(x))_{n \in \mathbb{N}} \in U^{\mathbb{N}}$ of H (for the inner product $\langle \cdot, \cdot \rangle_{H,H}$) of associated eigenvectors. Let's name them $(\rho_n)_{n \in \mathbb{N}}$. The theorem also holds that the sequence $(\frac{\rho_n}{\sqrt{\lambda_n}})$ is a Hilbert basis of U for the inner product $a(\cdot, \cdot)$.

If ρ is solution to (1) then there exists a sequence $(t \mapsto \nu_n(t))_{n \in \mathbb{N}}$ of functions of $H^2([0, T])$ such that:

$$\rho(t, x) = \sum_{n \in \mathbb{N}} \nu_n(t) \rho_n(x)$$

The values of ρ_n and of ν_n must be found.

$$-\mu \Delta u = \lambda u$$

with $\lambda > 0$ So

$$u(x) = A \cos \left(\sqrt{\frac{\lambda}{\mu}} x \right) + B \sin \left(\sqrt{\frac{\lambda}{\mu}} x \right)$$

with $A, B \in \mathbb{R}^I$

Moreover, we know that $u(0) = Pu(0)$ and that P is irreducible and thus has an eigenspace associated to the eigenvalue 1 of dimension 1. Let us denote V a vector of norm 1 of this space. As each V_i is positive (given the ergodic theorem) we have $\|V\|_1 = \sum_i V_i = 1$. Then, $A = \alpha V$ with $\alpha \in \mathbb{R}$

With $u_i(0) = u_i(L)$ we see that :

$$B \sin \left(\sqrt{\frac{\lambda}{\mu}} L \right) = A \left[1 - \cos \left(\sqrt{\frac{\lambda}{\mu}} L \right) \right]$$

If $\sqrt{\frac{\lambda}{\mu}} L \notin \pi\mathbb{Z}$ then :

$$B = \alpha V \frac{(1 - \cos(\sqrt{\frac{\lambda}{\mu}} L))}{\sin(\sqrt{\frac{\lambda}{\mu}} L)}$$

Because $\sum_i \int_0^L \rho_i = 0$ we have :

$$\sum_i \left[A_i \sin \left(\sqrt{\frac{\lambda}{\mu}} L \right) + B_i \left(1 - \cos \left(\sqrt{\frac{\lambda}{\mu}} L \right) \right) \right] = 0$$

i.e

$$\alpha \left(1 - \cos \left(\sqrt{\frac{\lambda}{\mu}} L \right) \right)^2 = -\alpha \sin \left(\sqrt{\frac{\lambda}{\mu}} L \right)^2$$

therefore, as $\alpha \neq 0$ because $u \neq 0$, we have :

$$(1 - \cos(\sqrt{\frac{\lambda}{\mu}} L))^2 = -\sin^2(\sqrt{\frac{\lambda}{\mu}} L)$$

We end up with the following equation :

$$\cos \left(\sqrt{\frac{\lambda}{\mu}} L \right) = 1$$

i.e $\sqrt{\frac{\lambda}{\mu}} L \in 2\pi\mathbb{Z}$, contradiction.

Therefore, $\sqrt{\frac{\lambda}{\mu}} L \in \mathbb{N}^* \pi$

$\sqrt{\frac{\lambda}{\mu}} L = n\pi$ with $n \in \mathbb{Z}^*$.

- If $n \in 2\mathbb{N}^*$ then the equation $\sum_i \int_0^L \rho_i = 0$ is trivial, and there is no constraint on the B_i .
- If $n \in 2\mathbb{N}^* + 1$ then we have :

$$\sum_i B_i = 0$$

and :

$$A(1 - \cos(\sqrt{\frac{\lambda}{\mu}} L)) = 2A = 0$$

so

$$\alpha = 0$$

We have eventually found all the eigen vectors of the operator a . These are the elements of the following sets :

$$\begin{aligned}
& - \{x \mapsto V \cos(\frac{2n\pi x}{L})\}_{n \in \mathbb{N}^*} \\
& - \{x \mapsto e_1 \sin(\frac{2n\pi x}{L})\}_{n \in \mathbb{N}^*} \\
& - \dots \\
& - \{x \mapsto e_I \sin(\frac{2n\pi x}{L})\}_{n \in \mathbb{N}^*} \\
& - \left\{x \mapsto \sum_{i=1}^I \gamma_{2n+1}^i e_i \sin(\frac{(2n+1)\pi x}{L}) \mid \sum_{i=1}^I \gamma_{2n+1}^i = 0\right\}_{n \in \mathbb{N}^*}
\end{aligned}$$

As shown before, these vectors form a Hilbert basis of U equipped with the inner product $a(.,.)$ up to normalization. Therefore we can decompose ρ in this basis : there exist $\mathcal{C}^1([0, T], \mathbb{R})$ functions $\alpha_{2n}, \beta_{2n}^i, \gamma_{2n+1}^i \forall n \in \mathbb{N}^*, i \in \{1, \dots, I\}$ such that

$$\rho(x, t) = \sum_{n \in \mathbb{N}^*} \left[\alpha_{2n}(t) V \cos\left(\frac{2n\pi x}{L}\right) + \sum_{i=1}^I \beta_{2n}^i(t) e_i \sin\left(\frac{2n\pi x}{L}\right) + \sum_{i=1}^I \gamma_{2n+1}^i(t) e_i \sin\left(\frac{(2n+1)\pi x}{L}\right) \right]$$

with $\sum_{i=1}^I \gamma_{2n+1}^i(t) = 0 \forall t$

Therefore for all $n \in \mathbb{N}^*, i \in \{1, \dots, I\}$, $\alpha_{2n}, \beta_{2n}^i, \gamma_{2n+1}^i$ satisfy the respective equations

$$\begin{cases} \alpha'_{2n}(t) = \frac{-4n^2\pi^2\mu}{L^2} \alpha_{2n}(t) \\ \beta_{2n}^{i'}(t) = \frac{-4n^2\pi^2\mu}{L^2} \beta_{2n}^i(t) \\ \gamma_{2n+1}^{i'}(t) = \frac{-(2n+1)^2\pi^2\mu}{L^2} \gamma_{2n+1}^i(t) \end{cases} \quad \text{for all } t \in [0, T]$$

Therefore,

$$\begin{cases} \alpha_{2n}(t) = \alpha_{2n}(0) e^{\frac{-4n^2\pi^2\mu}{L^2} t} \\ \beta_{2n}^i(t) = \beta_{2n}^i(0) e^{\frac{-4n^2\pi^2\mu}{L^2} t} \\ \gamma_{2n+1}^i(t) = \gamma_{2n+1}^i(0) e^{\frac{-(2n+1)^2\pi^2\mu}{L^2} t} \end{cases} \quad \text{for all } t \in [0, T]$$

Furthermore, as for all $n \in \mathbb{N}^*, i \in \{1, \dots, I\}$ we have :

$$\rho_0(x) = \sum_{n \in \mathbb{N}^*} \left[\alpha_{2n}(0) V \cos\left(\frac{2n\pi x}{L}\right) + \sum_{i=1}^I \beta_{2n}^i(0) e_i \sin\left(\frac{2n\pi x}{L}\right) + \sum_{i=1}^I \gamma_{2n+1}^i(0) e_i \sin\left(\frac{(2n+1)\pi x}{L}\right) \right]$$

with for all $n \in \mathbb{N}^*$:

$$\begin{cases} \alpha_{2n}(0) = \frac{2}{L\|V\|_2^2} \int_0^L \rho_0(x) \cdot V \cos\left(\frac{2n\pi x}{L}\right) dx & \text{for a certain } V_j \neq 0 \\ \beta_{2n}^i(0) = \frac{2}{L} \int_0^L \rho_0(x) \cdot e_i \sin\left(\frac{2n\pi x}{L}\right) dx \\ \gamma_{2n+1}^i(0) = \frac{2}{L} \int_0^L \rho_0(x) \cdot e_i \sin\left(\frac{(2n+1)\pi x}{L}\right) dx \end{cases}$$

The coefficients α, β and γ are imposed by the initial condition. We have therefore found the unique solution to (1).

3 Convergence of the cellular model

3.1 Numerical scheme of the PDE

We are looking for an approximation of the exact solution ρ . We denote $\rho_j^n = \rho(t_n, x_j)$ and $\rho_h^n = (\rho_0^n, \rho_1^n, \dots, \rho_{N-1}^n)^T$

In this section, we impose that P must be irreducible, so that there exists a unique vector V such that :

$$\|V\|_1 = 1, V_i \geq 0 \forall i \text{ and } PV = V$$

Using the expression of the discrete Laplacian and time derivative, we end up with this scheme :

$$\begin{cases} \frac{v_j^{n+1} - v_j^n}{\Delta_t} - \mu \frac{v_{j+1}^n - 2v_j^n + v_{j-1}^n}{\Delta_x^2} = 0, & \forall j \in \{1, \dots, N-1\}, n \in \{0, \dots, M-1\}, \\ \sum_{i=1}^I \left(\frac{v_{1,i}^n - v_{0,i}^n}{\Delta_x} - \frac{v_{0,i}^n - v_{N-1,i}^n}{\Delta_x} \right) = 0, & \forall n \in \{0, \dots, M-1\}, \\ v_0^n = v_N^n, & \forall n \in \{0, \dots, M\}, \\ Pv_0^n = v_0^n, & \forall n \in \{0, \dots, M\}. \end{cases}$$

We can show that it is equivalent to :

$$\forall n \in \{0, \dots, M-1\}$$

$$\begin{cases} v_j^{n+1} = \nu(v_{j-1}^n + v_{j+1}^n) + (1-2\nu)v_j^n & \forall j \in \{1, \dots, N-2\} \\ v_{N-1}^{n+1} = \nu(v_{N-2}^n + v_0^n) + (1-2\nu)v_{N-1}^n \\ v_0^n = \alpha^n V \text{ with } \alpha^n = \frac{1}{2}(\sum_{i=1}^I v_{1,i}^n + v_{N-1,i}^n) \end{cases}$$

3.2 Consistency, Stability and Convergence Analysis

We now show that the cellular model defined in the previous section is consistent and stable in the discrete L^2 norm, and therefore convergent toward the continuous diffusion equation.

Discrete notation. For each time index n , we collect the discrete unknowns in a vector

$$v_h^n = (v_0^n, v_1^n, \dots, v_{N-1}^n)^T \in \mathbb{R}^{IN},$$

where each component $v_j^n \in \mathbb{R}^I$ represents the concentrations of the I layers at node x_j . The discrete L^2 inner product and norm are defined by

$$(u_h, v_h)_h = \Delta_x \sum_{j=0}^{N-1} (u_j^T v_j), \quad \|u_h\|_h = (u_h, u_h)_h^{1/2}.$$

Matrix form of the scheme. The fully discrete explicit scheme can be written compactly as

$$v_h^{n+1} = Qv_h^n, \tag{1}$$

where Q is a block matrix of size $IN \times IN$ that includes:

- the tridiagonal coupling due to horizontal diffusion with coefficients $(\nu, 1-2\nu, \nu)$,
- the boundary connections involving the matrix P .

Consistency. We haven't shown the consistency at this point. Let ρ be a sufficiently exact solution of the continuous diffusion equation. We insert ρ into the discrete scheme to obtain the local truncation (consistency) error:

$$\tau_j^n = \frac{\rho(x_j, t_{n+1}) - \rho(x_j, t_n)}{\Delta_t} - \mu \frac{\rho(x_{j+1}, t_n) - 2\rho(x_j, t_n) + \rho(x_{j-1}, t_n)}{\Delta_x^2}, \quad j = 2, \dots, N-2.$$

For the interior nodes, the Taylor expansion of ρ in t and x around (x_j, t_n) gives

$$\tau_j^n = \frac{\Delta_t}{2} \partial_{tt} \rho(x_j, t_n) - \frac{\mu \Delta_x^2}{12} \partial_{x^4} \rho(x_j, t_n) + \mathcal{O}(\Delta_t^2 + \Delta_x^4), \quad (*)$$

But we haven't shown that this also holds for the boundary nodes. τ_0^n, τ_1^n and τ_{N-1}^n are subject to the same approximation.

Let us assume that $(*)$ holds for every value of j . Hence for a certain $C > 0$ independent of N and M :

$$\|\tau_j^n\|_2 \leq C(\Delta_t + \Delta_x^2) \quad \text{for all } n \in \{0, \dots, M-1\}, j \in \{0, \dots, N-1\}.$$

Therefore,

$$\sup_{0 \leq n \leq M} \|\tau_h^n\|_h \leq C(\Delta_t + \Delta_x^2),$$

and the scheme is **consistent of order** $(1, 2)$.

so the scheme is first order in time and second order in space. This proves the consistency of the method.

Stability in the discrete L^2 norm. It is easy to establish that the matrix Q satisfies

$$\|Q^n\|_2 \leq 1 \quad \text{for all } n \geq 0,$$

provided that the classical CFL condition

$$0 < \nu = \frac{\mu \Delta_t}{\Delta_x^2} \leq \frac{1}{2} \quad (2)$$

is fulfilled. The scheme is stable.

Convergence. Let us define the global error $e_h^n = v_h^n - \rho_h^n$. From the scheme and the consistency relation, we obtain

$$e_h^{n+1} = Q e_h^n + \Delta_t \tau_h^n.$$

Iterating and applying the discrete norm gives

$$\|e_h^{n+1}\|_h \leq \|Q^{n+1}\| \|e_h^0\|_h + \Delta_t \sum_{m=0}^n \|Q^{n-m}\| \|\tau_h^m\|_h.$$

Using $\|Q^k\| \leq 1$ and $e_h^0 = 0$,

$$\|e_h^{n+1}\|_h \leq T \sup_m \|\tau_h^m\|_h \leq C T (\Delta_t + \Delta_x^2).$$

Thus, if it is consistent, the scheme is convergent of order $\mathcal{O}(\Delta_t + \Delta_x^2)$ in discrete L^2 norm.

4 Numerical study of the convergence

Let us examine the convergence of the cellular model for three simple cases: a three-layer basin, with initial functions corresponding to the system's own functions. We will limit ourselves to cases where the coefficients $\alpha_{2k}, \beta_{2k}^i, \gamma_{2k+1}^i$ are zero if and only if $k = 1$. Let us take the simple examples of initial functions corresponding to the system's eigen values. That is functions of the same form as ρ in section 2, with :

- $\alpha_{2k} = \delta_{k,1}, \beta_{2k}^i = 0, \gamma_{2k+1}^1 = \gamma_{2k+1}^2 = \gamma_{2k+1}^3 = 0.$
- $\alpha_{2k} = 0, \beta_{2k}^i = \delta_{k,1}, \gamma_{2k+1}^1 = \gamma_{2k+1}^2 = \gamma_{2k+1}^3 = 0.$
- $\alpha_{2k} = 0, \beta_{2k}^i = \delta_{k,1}, \gamma_3^1 = \gamma_3^2 = 1 \text{ and } \gamma_3^3 = -2\gamma_{2k+1}^1 = \gamma_{2k+1}^2 = \gamma_{2k+1}^3 = 0 \forall k \neq 0.$

5 Mixing Optimization : Optimal Matrix

I have developed a numerical algorithm to model algal growth in a water basin. The model is based on the introduction of a source term into the discretization of the advection–diffusion equation with an upwind discretization of the advection term.

Let us first introduce the numerical scheme. Far from the paddle wheel, the PDE discretization leads to :

$$\forall n \in \{0, \dots, M-1\}$$

$$\frac{v_j^{n+1} - v_j^n}{\Delta t} - D \frac{v_{j+1}^n - 2v_j^n + v_{j-1}^n}{\Delta x^2} + a \frac{v_j^n - v_{j-1}^n}{\Delta x} = 0$$

and the numerical scheme is defined by :

$$\begin{cases} v_j^{n+1} = (\kappa + \nu)v_{j-1}^n + (1 - 2\nu - \kappa)v_j^n + \nu v_{j+1}^n & \forall j \in \{2, \dots, N-2\} \\ v_{N-1}^{n+1} = (\kappa + \nu)v_{N-2}^n + (1 - 2\nu - \kappa)v_{N-1}^n + \nu P v_0^n \\ v_0^{n+1} = (\kappa + \nu)v_{N-1}^n + (1 - 2\nu - \kappa)P v_0^n + \nu v_1^n \\ v_1^{n+1} = (\kappa + \nu)P v_0^n + (1 - 2\nu - \kappa)v_1^n + \nu v_2^n \end{cases}$$

Let us now introduce the source term. Each layer i corresponds to a source term μ_i . Let $\mu = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_I \end{pmatrix} \in \mathbb{R}^I$.

$$\begin{cases} v_j^{n+1} = (\kappa + \nu)v_{j-1}^n + (1 - 2\nu - \kappa + \mu\Delta t)v_j^n + \nu v_{j+1}^n & \forall j \in \{2, \dots, N-2\} \\ v_{N-1}^{n+1} = (\kappa + \nu)v_{N-2}^n + (1 - 2\nu - \kappa + \mu\Delta t)v_{N-1}^n + \nu P v_0^n \\ v_0^{n+1} = (\kappa + \nu)v_{N-1}^n + (1 - 2\nu - \kappa + \mu\Delta t)P v_0^n + \nu v_1^n \\ v_1^{n+1} = (\kappa + \nu)P v_0^n + (1 - 2\nu - \kappa + \mu\Delta t)v_1^n + \nu v_2^n \end{cases}$$

This additional term modifies the Courant–Friedrichs–Lewy (CFL) condition and, consequently, the value of the time step Δt . To converge, the numerical scheme must verify :

$$\Delta t \leq \frac{\Delta x^2}{2\mu + a\Delta x}$$

5.1 introduction of a biomass source term

In this report, we present first numerical results illustrating the total biomass in the basin after a growth period of $T = 100$ s. According to [1], the algae follow the Han growth model, which is of Haldane if the dynamic of the open state is at its steady state. The light intensity follows a Beer-lambert law [2]:

$$I = I_s e^{-\epsilon z}$$

With $I_s = 1500 \text{ m}^{-2} \mu\text{mol s}^{-1}$

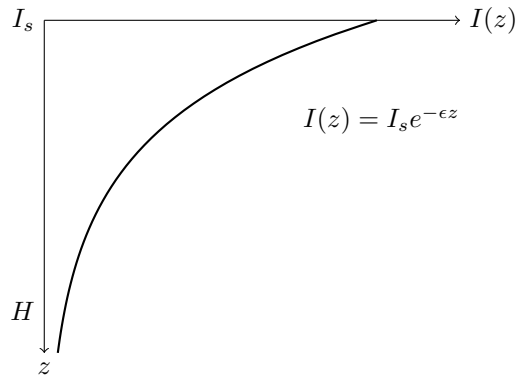


Figure 1: Exponential attenuation of light intensity with depth according to the Beer–Lambert law.

The ϵ light extinction coefficient will be computed once and for all, assuming that the light intensity at depth H is 1% of the surface light intensity I_s . We have :

$$I(H) = \frac{I_s}{100} = I_s \exp(-\epsilon H)$$

Therefore,

$$\epsilon = \frac{\ln(100)}{H} \approx 4.6$$

And for the values of the constants involved in the computing of μ :

$$k_d = 2.99 \times 10^{-5},$$

$$k_r = 4.8 \times 10^{-5} \text{ s}^{-1},$$

$$k_h = 3.64 \times 10^{-5},$$

$$\tau = 6.849 \text{ s},$$

$$\sigma_H = 2.9 \times 10^{-4} \text{ m}^2 \mu\text{mol}^{-1}$$

We have

$$A = \frac{1}{\frac{k_d}{k_r} \tau (\sigma_H I)^2 + \tau \sigma_H I + 1}$$

And

$$\mu = A I k_h \sigma_H = 1.78 \cdot 10^{-4}$$

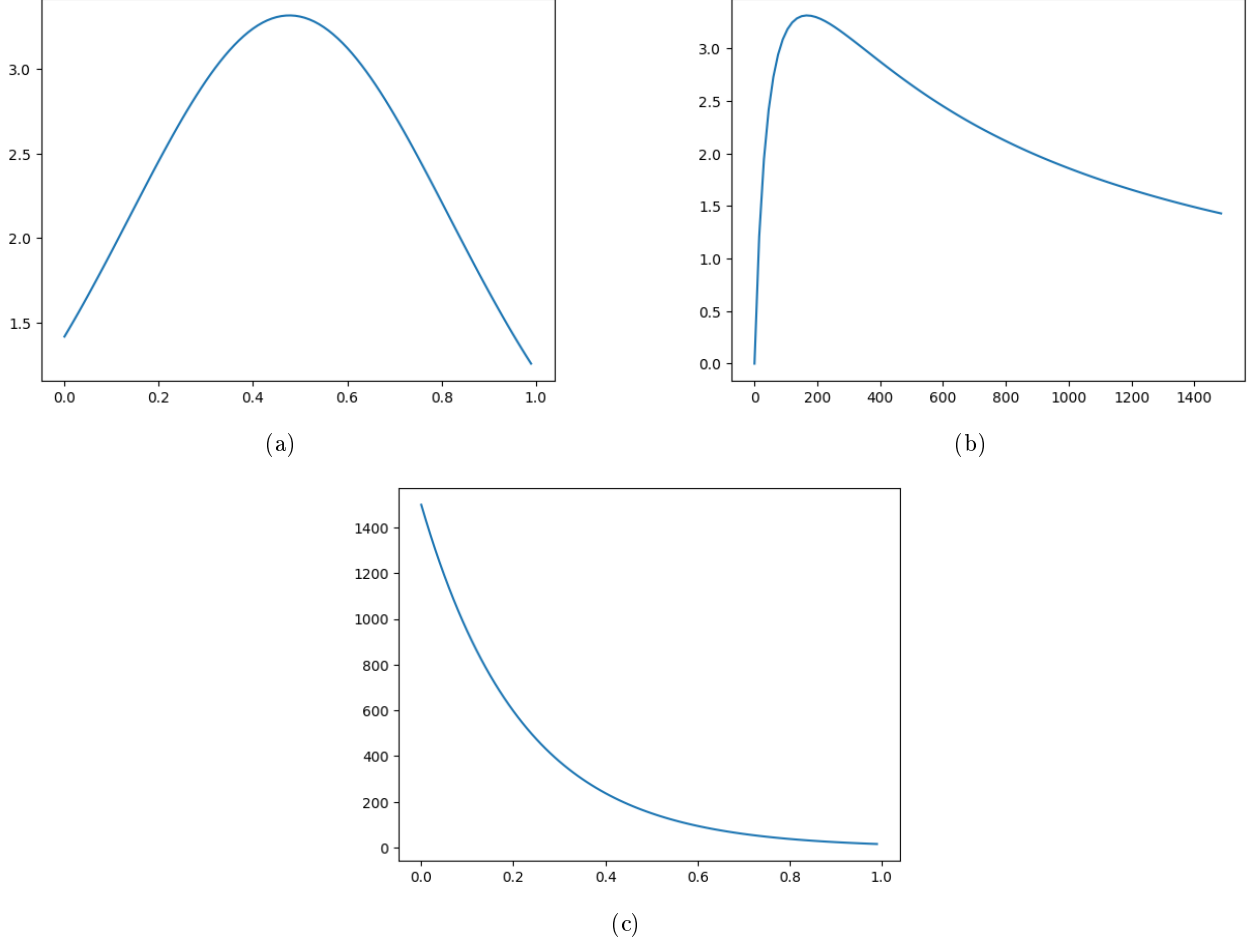


Figure 2: (a) $\mu (d^{-1})$ as a function of z ; (b) $\mu (d^{-1})$ as a function of I ; (c) I as a function of z

The simulations are performed for a single layer, and therefore without any mixing effects. The only varying parameter is the water velocity u . For a discretization of 500 points, we impose a uniform initial concentration and an initial total biomass of 50 Kg. After an hour of growth, we get the following total biomass table for different current speed values.

Current speed (m.s ⁻¹)	Total biomass (kg)
0	52.8041
0.1	52.6224
0.2	52.6222
0.3	52.6221
0.5	52.6221
1	52.6221

Table 1: Total biomass ($T = 3600$ s), as a function of current speed

It seems, that the biomass grows less as the current increases. One must take into account that Δ_t , the time step of the numerical scheme depends on the value of the speed u . This follows from the upwind scheme CFL condition.

6 Discrete Optimization over Permutation Matrices

6.1 Motivation

Before addressing the general optimization problem over stochastic or bistochastic matrices, we first restrict the admissible set to permutation matrices.

This restriction is motivated by both computational and physical considerations. From a computational perspective, the finite cardinality of the set of permutation matrices allows for exhaustive search for small system sizes. From a physical point of view, permutation matrices correspond to deterministic transport of biomass between layers, as opposed to diffusive mixing.

6.2 Problem Definition

The objective of the algorithm is to optimize a mixing matrix P applied to a discretized basin, in order to maximize the total biomass produced by the system. In this section, we restrict P to the set of permutation matrices:

$$\mathcal{P}_N = \{P \in \{0, 1\}^{N \times N} \mid P\mathbf{1} = \mathbf{1}, P^\top \mathbf{1} = \mathbf{1}\}.$$

Given an initial biomass distribution $B(0)$, the system is evolved over a fixed time horizon T using a time-discretized dynamical model (to be detailed in Section ??). The objective function is defined as the total biomass at final time:

$$J(P) = B_{\text{tot}}(T).$$

The optimization problem reads:

$$\max_{P \in \mathcal{P}_N} J(P).$$

6.3 Exhaustive Search Algorithm

For small values of N , the optimization problem can be solved by exhaustive enumeration of all permutation matrices.

The number of such matrices is $N!$, which rapidly becomes computationally prohibitive as N increases. In practice, exhaustive search is limited to:

$$N \leq 7.$$

The algorithm proceeds as follows:

1. Generate all permutation matrices $P \in \mathcal{P}_N$;
2. For each P , simulate the system dynamics up to time T ;
3. Compute the objective value $J(P)$;
4. Select the matrix maximizing J .

6.4 Tested Configurations

The total initial biomass is fixed:

$$B_{\text{tot}}(0) = 50 \text{ Kg.}$$

The growth time is fixed to:

$$T = [\text{TO COMPLETE}],$$

and the flow velocity is set to:

$$u = 0.1 \text{ m s}^{-1}$$

300 seconds of growth therefore correspond to approximately 6 laps.

To assess the influence of initial conditions, several configurations are considered :

- biomass entirely localized in a single layer, while all other layers are initialized with zero biomass. This leads to N distinct initial configurations.
- Uniform distribution of the biomass.

6.5 Results for Small Systems

6.5.1 Case $N = 3$

The growth rates associated with each layer are given in Table 2.

Layer	$\mu \text{ (s}^{-1}\text{)}$	Light intensity
Layer 1	1.4198×10^{-5}	1500
Layer 2	3.0538×10^{-5}	323.72
Layer 3	2.8777×10^{-5}	69.86

Table 2: Growth rates and light intensities for $N = 3$

For each initial condition, the optimal permutation matrix is determined. The results are summarized below.

- Configuration 1 (biomass initially in layer #1):

Optimal matrix:

$$\mathbf{P}_1 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

Final biomass:

$$B_{\text{tot}} = 50.369 \text{ Kg}$$

This matrix induces a permutation of layers which induces the allocation of layer 1 in position 2.

- Configuration 2 (biomass initially in layer #2):

Optimal matrix:

$$\mathbf{P}_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Final biomass:

$$B_{\text{tot}} = 50.461 \text{ Kg}$$

Here, no mixing appears to be the optimal strategy.

- Configuration 3 (biomass initially in layer #3):

Optimal matrix:

$$\mathbf{P}_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$

Final biomass:

$$B_{\text{tot}} = 50.447 \text{ Kg}$$

This configuration corresponds to the permutation of layers 2 and 3.

- Configuration 4 : uniform distribution

Optimal matrix:

$$\mathbf{P}_4 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Final biomass:

$$B_{\text{tot}} = 50.370 \text{ Kg}$$

As in case 2, no mixing appears to be the optimal strategy.

6.6 Discussion

The results reveal a strong dependence of the optimal strategy on the initial configuration.

When biomass is initially located in a layer with high growth rate, the identity permutation (i.e., no mixing) is optimal. We observe the same phenomenon for a uniform initial distribution. In contrast, when biomass is initially located in less favorable layers, the optimal permutations transport biomass toward layers with higher growth rates.

More generally, the optimal permutation matrices tend to concentrate biomass in the layers maximizing the local growth coefficient μ .

This behavior is consistent with the physical interpretation of the system: the mixing strategy aims at maximizing the effective light exposure experienced by the biomass over time.

These observations will serve as a basis for the analysis of more general mixing strategies in the following sections.

Here is a graphic representing the total algae mass in each layer as a function of time, for configuration 3 and using the mixing matrix $\mathbf{P}_3$.

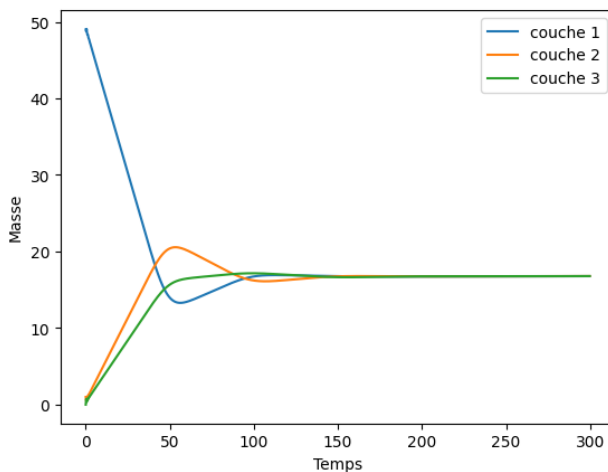


Figure 3: biomass in each layer through time

We notice that the algae concentration converges to a limit in each layer. As supposed earlier, the vector of concentrations in the column corresponding to the paddle converges to an eigen vector of the matrix. This phenomenon has been observed with different initial conditions, and with N varying from 3 to 7.

7 Structural Insights from Permutation-Based Optimization

7.1 Emerging Patterns in Optimal Permutations

The exhaustive optimization over permutation matrices reveals consistent structural patterns across all tested configurations and system sizes.

In all cases, the optimal permutations exhibit a strong tendency to transport biomass toward layers associated with higher growth rates. More precisely, the permutation matrices maximizing the objective function $J(P)$ tend to map a large fraction of the biomass trajectories toward the subset of layers where the growth coefficient μ is maximal.

This behavior is observed regardless of the initial condition, although the optimal permutation depends on the initial localization of the biomass.

7.2 Interpretation in Terms of Biomass Trajectories

A permutation matrix defines a deterministic mapping between layers. Iterating such a matrix over time induces periodic trajectories for each biomass element.

Therefore, instead of interpreting the mixing process as a spatial redistribution, it is natural to adopt a Lagrangian viewpoint and consider the temporal trajectories experienced by individual biomass particles.

Under this interpretation, a permutation matrix generates cyclic trajectories:

$$i \mapsto P(i) \mapsto P^2(i) \mapsto \dots$$

The growth of each biomass element is then governed by the sequence of light intensities encountered along its trajectory.

7.3 Connection with the Flashing Light Effect

The observed structure of optimal permutations can be directly related to the flashing light effect.

As discussed previously, the photosynthetic response is nonlinear with respect to light intensity. In particular:

- high light intensities may lead to saturation or photoinhibition,
- low light intensities result in suboptimal growth,
- intermittent exposure can enhance overall efficiency.

Permutation-based mixing induces structured temporal alternations between different light environments. In particular, optimal permutations tend to maximize the time spent in high-growth layers while still maintaining temporal variability.

This suggests that the optimization process naturally balances two competing effects:

- maximizing exposure to favorable conditions,
- avoiding constant exposure that would reduce efficiency.

7.4 Extremal Nature of Optimal Strategies

An important observation is that permutation matrices correspond to extremal points of the set of bistochastic matrices.

Indeed, any bistochastic matrix can be written as a convex combination of permutation matrices:

$$P = \sum_k \lambda_k P_k, \quad \lambda_k \geq 0, \quad \sum_k \lambda_k = 1.$$

From this perspective, stochastic or bistochastic mixing can be interpreted as a probabilistic combination of deterministic trajectories.

The numerical results suggest that such averaging effects are detrimental to the objective function. Indeed, spreading biomass over multiple trajectories tends to reduce the temporal contrast in light exposure, thereby weakening the flashing light effect.

In contrast, permutation matrices preserve fully coherent trajectories, which appear to be more favorable for growth.

7.5 Heuristic Explanation of Optimality

The previous observations lead to the following heuristic explanation:

Optimal mixing strategies maximize the temporal variability of light exposure while ensuring frequent visits to high-growth regions.

Permutation matrices naturally achieve this by:

- enforcing structured, periodic trajectories,
- avoiding diffusive spreading of biomass,
- concentrating exposure in favorable layers.

This provides a qualitative explanation for the empirical observation that optimal solutions are consistently found among permutation matrices.

7.6 Implications for the Continuous Optimization Problem

These structural insights strongly suggest that the relaxation of the optimization problem to larger sets (stochastic or bistochastic matrices) may not lead to better solutions.

Instead, it is expected that optimal solutions remain located near the extremal points of the admissible set.

This hypothesis will be investigated in the following sections using gradient-based optimization methods over continuous matrix spaces.

8 Mixing Optimization : Optimal Stochastic Mixing-Matrix

Now that we have found a way to determine the best permutation matrix for the water pond, let us now focus on a more general case : the column stochastic matrices. The initial problem is indeed defined with P a $N \times N$ column-stochastic matrix. To optimize on this infinite space, we will have to use a differentiable program. We have chosen the torch python library to do so. In the update fonction, we have to convert every numpy list and array into a torch variable. This will allow us to compute torch differentiation with these variables. Furthermore, to process the matrix constraints (non negative coefficients and every column must sum 1), we will use the softmax function, in order to differentiate with respect an unconstrained matrix Q .

The objective function remains the same.

8.1 Optimal stochastic matrix

Let us now use the original objective function, i.e the final biomass after a given growth period T . As before, the growth time remains $T = 300s$ and the initial conditions are that the total biomass is in layer 1. The algorithm complexity is a linear function of the number of layers. Therefore, it allows us to divide the water pond in 10 layers.

8.1.1 optimizer parameters

The Adam optimizer (Adaptive Moment Estimation) combines momentum-based gradient descent with per-parameter adaptive learning rates. At each iteration t , it maintains an exponential moving average of the gradient m_t (first moment) and of the squared gradient v_t (second moment), defined as

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t, \quad v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2,$$

where g_t denotes the gradient of the loss with respect to the parameters. These moment estimates are corrected for initialization bias as

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}.$$

The parameter update is then given by

$$\theta_{t+1} = \theta_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \varepsilon},$$

where α is the global learning rate, while β_1 and β_2 control the exponential decay rates of the first and second moments, respectively, governing the trade-off between stability and responsiveness of the optimization process.

In order to choose an appropriate learning rate for the Adam optimization algorithm, we performed a learning rate finder experiment. The principle consists in launching the optimization with different α values, and check which one converges the quickest.

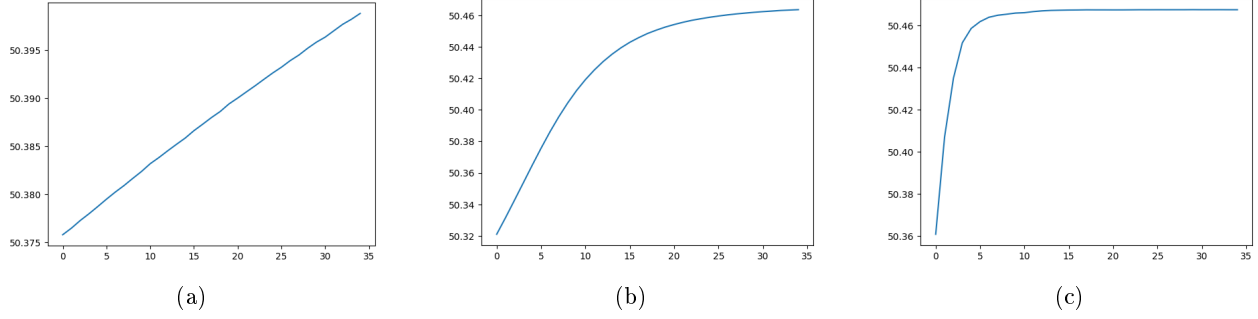


Figure 4: (a) $\alpha = 0.01$; (b) $\alpha = 0.1$; (c) $\alpha = 0.9$

It is clear that the best learning rate is $\alpha = 0.9$. It reaches a plateau after about 12 iterations. Based on this observation, we can arbitrarily bond the number of iterations by 12.

8.1.2 optimal stochastic matrix

For 10 layers, the approximate optimal matrix is the following.

$$\begin{pmatrix} 0.000 & 0.003 & 0.010 & 0.000 & 0.000 & 0.000 & 0.001 & 0.001 & 0.001 & 0.004 \\ 0.000 & 0.011 & 0.001 & 0.020 & 0.000 & 0.000 & 0.000 & 0.007 & 0.004 & 0.006 \\ 0.000 & 0.009 & 0.008 & 0.003 & 0.000 & 0.000 & 0.000 & 0.005 & 0.004 & 0.004 \\ 0.001 & 0.394 & 0.373 & 0.080 & 0.000 & 0.000 & 0.002 & 0.112 & 0.151 & 0.082 \\ 0.003 & 0.285 & 0.251 & 0.072 & 0.012 & 0.000 & 0.504 & 0.073 & 0.193 & 0.051 \\ 0.992 & 0.095 & 0.076 & 0.443 & 0.985 & 0.999 & 0.345 & 0.511 & 0.331 & 0.599 \\ 0.003 & 0.087 & 0.140 & 0.371 & 0.001 & 0.000 & 0.142 & 0.283 & 0.268 & 0.203 \\ 0.000 & 0.103 & 0.131 & 0.005 & 0.001 & 0.000 & 0.005 & 0.001 & 0.041 & 0.045 \\ 0.000 & 0.007 & 0.005 & 0.006 & 0.000 & 0.000 & 0.000 & 0.001 & 0.001 & 0.002 \\ 0.000 & 0.007 & 0.006 & 0.000 & 0.000 & 0.000 & 0.000 & 0.005 & 0.005 & 0.002 \end{pmatrix}$$

As seen before, the line 6 has the largest coefficients because layer 6 has the highest growth coefficient. The optimal matrix allocates most of the biomass in this layer. But stochastic matrices aren't physically relevant. Indeed, in terms of water volume, a real-life paddle wheel should allocate to each layer exactly the volume of one layer. Such a wheel could only be represented by a bistochastic matrix.

9 Optimal bistochastic matrix

Using the same solver than before, we only add a penalty to the objective function to impose that the sum on each line of the mixing matrix should equal 1. After launching the program for a 10 layers pond, we end up with the following matrix.

$$\begin{bmatrix} 0.1727 & 0.0401 & 0.1280 & 0.1118 & 0.0686 & 0.0839 & 0.0447 & 0.2331 & 0.0285 & 0.0885 \\ 0.0422 & 0.2675 & 0.0660 & 0.1765 & 0.0297 & 0.0281 & 0.1194 & 0.1010 & 0.0907 & 0.0790 \\ 0.2268 & 0.0297 & 0.1286 & 0.0654 & 0.0687 & 0.0290 & 0.2182 & 0.0791 & 0.1469 & 0.0075 \\ 0.1335 & 0.1178 & 0.0531 & 0.0894 & 0.1766 & 0.0053 & 0.0866 & 0.0164 & 0.0864 & 0.2348 \\ 0.2559 & 0.0255 & 0.3115 & 0.0160 & 0.0122 & 0.0688 & 0.0067 & 0.1059 & 0.1850 & 0.0123 \\ 0.0238 & 0.3348 & 0.0341 & 0.0567 & 0.2197 & 0.0715 & 0.0415 & 0.0387 & 0.1417 & 0.0375 \\ 0.0467 & 0.0217 & 0.0830 & 0.3417 & 0.0783 & 0.0183 & 0.2618 & 0.0856 & 0.0249 & 0.0380 \\ 0.0410 & 0.0510 & 0.0688 & 0.0451 & 0.2185 & 0.1392 & 0.0301 & 0.2752 & 0.1097 & 0.0213 \\ 0.0356 & 0.0181 & 0.0586 & 0.0700 & 0.1100 & 0.5202 & 0.0761 & 0.0173 & 0.0743 & 0.0198 \\ 0.0218 & 0.0936 & 0.0683 & 0.0274 & 0.0177 & 0.0357 & 0.1148 & 0.0476 & 0.1119 & 0.4613 \end{bmatrix}$$

This mixing matrix corresponds to a final biomass (after $T = 300s$ of growth) of $B_{\text{tot}} = 50.369$ Kg.

The previous optimal permutation matrix algorithm allows us to compare the best permutation matrix to the bistochastic case, and to check if a bistochastic relaxation leads to a higher performance. In fact, what

was shown by the simulations, is that the optimal permutation matrices are at least a local maximum of the objective function on the set of bistochastic matrices. Indeed, launching the optimization algorithm with a starting point at the optimal permutation matrix leads to a plateau. This implies that no better bistochastic matrix near the permutation matrix can be found.

L'ensemble admissible étant convexe, ses points extrémaux (matrices de permutation) jouent un rôle particulier. Nos résultats numériques et analytiques suggèrent que la structure du fonctionnel favorise ces points extrémaux, en cohérence avec l'interprétation physique du flashing effect

10 Flashing effect

We have observed a phenomenon of higher growth as the basin's length lowers. For the same parameters, excepted the length L , the algae tend to grow more when the number of mixes increases. This can be explained by the fact that if each "layer", for the same total growing time T , gets to each position a larger number of times.

11 annexes

11.1 initial test

To check if the new solver is functional. In order to do so, let us consider a three layers basin, and an objective function which mixes these layers with respect to the matrix $P = \text{softmax}(Q)$ and returns the total biomass, after a pre-determined growing period. The growth coefficients will be arbitrarily set to 1 for layer 1, 2 for layer 2 and 3 for layer 3. Therefore, with an initial basin in which the total biomass is concentrated in layer 1, an optimal matrix should reallocate it in layer 3. Let us consider the results.

The optimal mixing matrix is:

$$\mathbf{P} = \begin{pmatrix} 0.003 & 0.50 & 0.26 \\ 0.007 & 0.30 & 0.10 \\ 0.99 & 0.20 & 0.64 \end{pmatrix}$$

The first layer is indeed moved to layer 3 by 99 percent.

11.2 3 layers comparison test

With an initial pond of 3 layers, in which the biomass is only in the first layer, the solver returns the following matrix :

$$\begin{pmatrix} 4.126 \times 10^{-5} & 3.704 \times 10^{-4} & 2.707 \times 10^{-4} \\ 9.961 \times 10^{-1} & 9.976 \times 10^{-1} & 9.848 \times 10^{-1} \\ 3.901 \times 10^{-3} & 2.034 \times 10^{-3} & 1.495 \times 10^{-2} \end{pmatrix}$$

this matrix corresponds to a situation in which the final biomass is 50.301Kg. This is approximately the same result than the permutation matrix case. The optimal permutation matrix was indeed :

$$\mathbf{M}_1 = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$

We can now begin the ADAM algorithm starting from this matrix and see if we obtain a better result. The final biomass result stays the same, while the optimisation returns a new matrix :

$$\begin{pmatrix} 7.768 \times 10^{-4} & 3.514 \times 10^{-4} & 3.007 \times 10^{-3} \\ 9.933 \times 10^{-1} & 9.974 \times 10^{-1} & 9.429 \times 10^{-1} \\ 5.955 \times 10^{-3} & 2.301 \times 10^{-3} & 5.411 \times 10^{-2} \end{pmatrix}$$

11.3 Newton method

Let us try to use another optimizer, based on the Newton method with conjugate gradient. This method is a second-order optimization technique used to find a local minimizer of a twice-differentiable function $f : \mathbb{R}^n \rightarrow \mathbb{R}$. At iteration k , the Newton direction p_k is defined as the solution of the linear system

$$\nabla^2 f(x_k) p_k = -\nabla f(x_k),$$

where $\nabla f(x_k)$ is the gradient and $\nabla^2 f(x_k)$ is the Hessian matrix.

For large-scale problems as ours, explicitly forming and inverting the Hessian matrix is computationally expensive. Instead, the Newton system can be solved approximately using the Conjugate Gradient (CG) method. The CG algorithm is particularly efficient when the Hessian is symmetric and positive definite, as it only requires matrix-vector products.

In the Newton-CG method, at each iteration we apply the Conjugate Gradient algorithm to compute an approximate solution p_k of the Newton system. The iterate is then updated as

$$x_{k+1} = x_k + \alpha_k p_k,$$

where α_k is determined by a line search procedure.

11.4 Computational complexity.

We analyze the computational complexity of the optimization procedure by first estimating the cost of a single evaluation of the objective function. The time step of the numerical scheme is given by

$$\Delta t = \text{CFL} \frac{\Delta x^2}{2\mu + a \Delta x}, \quad \text{with} \quad \Delta x = \frac{L}{n},$$

where n denotes the number of spatial discretization points, μ the diffusion coefficient, and a a characteristic advection velocity. Substituting Δx yields

$$\Delta t = \text{CFL} \frac{L^2}{n^2(2\mu + aL/n)} = \frac{\text{CFL} L^2}{2\mu n^2 + aLn}.$$

In our setting, typical parameter values are $\mu \approx 10^{-5}$, $n \approx 50$, $0.01 < a < 0.1$, and $L \approx 5$. Under these assumptions, the term aLn dominates $2\mu n^2$, leading to the approximation

$$\Delta t \sim \frac{\text{CFL} L}{a n}.$$

The objective function corresponds to the final state of a time-marching simulation over a time horizon T . The number of time steps is therefore of order $T/\Delta t$, and each time step involves $O(Nn)$ operations, where N denotes the number of layers. Consequently, the computational complexity of a single forward evaluation of the objective function is

$$\mathcal{O}\left(Nn \frac{T}{\Delta t}\right) = \mathcal{O}\left(\frac{a N n^2 T}{\text{CFL} L}\right).$$

Since the objective function is implemented in a differentiable manner using automatic differentiation, the backward pass requires propagating gradients through the entire time-marching scheme. The computational cost of the backward pass is of the same order as the forward pass, up to a constant factor, yielding

$$\mathcal{O}\left(\frac{a N n^2 T}{L}\right).$$

Finally, the full optimization procedure consists of K iterations of the Adam algorithm. Each iteration requires one forward evaluation of the objective function and one backward pass, while the update of the parameter matrix $P \in \mathbb{R}^{N \times N}$ is negligible in comparison. The overall computational complexity of the `solve` algorithm is therefore

$$\mathcal{O}\left(K \frac{a N n^2 T}{L}\right).$$

References

- [1] Liu-Di Lu, *Lagrangian Approaches for Modelling and Optimization of Hydrodynamic-Photosynthesis Coupling*, Sorbonne Université, LJLL, 2021
- [2] Oxford reference *Beer-Lambert Law*